Lecture 18 – Normal Forms of Context-Free Grammars

COSE215: Theory of Computation

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PLRG

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Recall

- A **context-free grammar (CFG)** is a 4-tuple:

$$G = (V, \Sigma, S, R)$$

  where
  - V: a finite set of **variables** (nonterminals)
  - Σ: a finite set of **symbols** (terminals)
  - $S \in V$: the **start variable**
  - $R \subseteq V \times (V \cup \Sigma)^*$: a set of **production rules**.

- How to **simplify** a CFG?

  Let's put it in **Chomsky normal form (CNF)**!

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3. Eliminating Unit Productions
   - Unit Pairs

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Chomsky Normal Form (CNF)

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Definition (Chomsky Normal Form)

A CFG G is in **Chomsky normal form (CNF)** if all productions are of the form for some $A, B, C \in V$ and $a \in \Sigma$:

$$A \to BC \qquad \text{OR} \qquad A \to a \qquad \text{OR} \qquad S \to \epsilon$$

where $B \neq S$ and $C \neq S$. And $S \to \epsilon$ is allowed only if $\epsilon \in L(G)$.

Consider the following CFG:

$$\begin{array}{lll} S \to 0ABC \mid 1B \mid BB & A \to ABB0 \mid C & C \to CC \mid \epsilon \\ & B \to 0B \mid 1 & D \to 1D \mid AA \end{array}$$

Is it possible to put this CFG in CNF? **Yes!**

$$\begin{array}{lll} S \to XS_1 \mid XB \mid YB \mid BB & A \to AA_1 \mid BA_2 & B \to XB \mid 1 \\ S_1 \to AB & A_1 \to BA_2 & X \to 0 \\ & A_2 \to BX & Y \to 1 \end{array}$$

Let's learn how to put a CFG in CNF!

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Eliminating ϵ-Productions

The following productions are called ϵ-**productions**:

$$A \to \epsilon$$

Is it possible to eliminate all ϵ-**productions** from a CFG?

No, if an empty word ϵ is in the language of the CFG (i.e., $\epsilon \in L(G)$), then we cannot generate the empty word without ϵ-productions.

However, we can eliminate all ϵ-productions from a CFG G to construct a new CFG G' such that:

$$L(G') = L(G) \setminus \{\epsilon\}$$

We can do it by following the steps below:

1. Find all **nullable variables**.
2. Construct a new CFG by **replacing** nullable variables with ϵ in **all combinations** and **removing** all ϵ-productions in production rules.

Nullable Variables

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Definition (Nullable Variables)

For a given CFG $G = (V, \Sigma, S, R)$, a variable $A \in V$ is **nullable** if

$$A \Rightarrow^* \epsilon$$

We can inductively define the set of **nullable variables**:

- **(Basis Case)** If $A \to \epsilon \in R$, then A is nullable.
- **(Induction Case)** If $A \to X_1 X_2 \cdots X_n \in R$ and $X_1, X_2, \ldots, X_n$ are all nullable, then A is nullable.

Eliminating ϵ-Productions – Example

Consider the following CFG:

$$
\begin{aligned}
S &\to 0ABC \mid 1B \mid BB \\
A &\to ABB0 \mid C \\
B &\to 0B \mid 1 \\
C &\to CC \mid \epsilon \\
D &\to 1D \mid AA
\end{aligned}
$$

1. Find all **nullable variables**: $\{A, C, D\}$
2. Construct a new CFG by **replacing** nullable variables with ϵ in **all combinations** and **removing** all ϵ-productions in production rules:

$$
\begin{aligned}
S &\to 0ABC \mid 0BC \mid 0AB \mid 0B \mid 1B \mid BB \\
A &\to ABB0 \mid BB0 \mid C \\
B &\to 0B \mid 1 \\
C &\to CC \mid C \\
D &\to 1D \mid 1 \mid AA \mid A
\end{aligned}
$$

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Eliminating Unit Productions

The following productions are called **unit productions**:

$$A \to B$$

Is it possible to eliminate **unit productions**?

Yes, we can do it by following the steps below:

1. Find all **unit pairs**.
2. Construct a new CFG by **adding** all possible non-unit productions of B to A for each unit pair (A, B).

Unit Pairs

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Definition (Unit Pairs)

For a given CFG $G = (V, \Sigma, S, R)$, a pair of variables $(A, B) \in V \times V$ is a **unit pair** if

$$A \Rightarrow^* B$$

We can inductively define the set of **unit pairs**:

- **(Basis Case)** (A, A) is a unit pair for all $A \in V$.
- **(Induction Case)** If (A, B) is a unit pair and $B \to C \in R$, then (A, C) is a unit pair.

Eliminating Unit Productions – Example

After eliminating ϵ-productions:

$$
\begin{aligned}
S &\to 0ABC \mid 0BC \mid 0AB \mid 0B \mid 1B \mid BB \\
A &\to ABB0 \mid BB0 \mid C \\
B &\to 0B \mid 1 \\
C &\to CC \mid C \\
D &\to 1D \mid 1 \mid AA \mid A
\end{aligned}
$$

1. Find all **unit pairs**:

$$\{(S,S),(A,A),(A,C),(B,B),(C,C),(D,D),(D,A),(D,C)\}$$

2. Construct a new CFG by **adding** all possible non-unit productions of B to A for each unit pair (A,B).

$$
\begin{aligned}
S &\to 0ABC \mid 0BC \mid 0AB \mid 0B \mid 1B \mid BB \\
A &\to ABB0 \mid BB0 \mid CC \\
B &\to 0B \mid 1 \\
C &\to CC \\
D &\to 1D \mid 1 \mid AA \mid ABB0 \mid BB0 \mid CC
\end{aligned}
$$

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Eliminating Useless Variables

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What are useless variables?

- **Non-generating variables**: Variables that cannot derive any word.
- **Unreachable variables**: Variables unreachable from the start variable.

Is it possible to eliminate **useless variables**?

Yes, we can do it by following the steps below:

1. Find all **generating variables**.
2. Find all **reachable variables**.
3. Construct a new CFG by **removing** all productions that contain non-generating variables or come from unreachable variables.

Generating Variables

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Definition (Generating Variables)

For a given CFG $G = (V, \Sigma, S, R)$, a variable $A \in V$ is a **generating variable** if for some $w \in \Sigma^*$,

$$A \Rightarrow^* w$$

We can inductively define the set of **generating variables**:

- **(Basis Case)** There is no basis case.
- **(Induction Case)** If $A \to \alpha \in R$ and α contains only symbols or generating variables, then A is a generating variable.

Reachable Variables

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Definition (Reachable Variables)

For a given CFG $G = (V, \Sigma, S, R)$, a variable $A \in V$ is a **reachable variable** if there exists a derivation:

$$S \Rightarrow^* \alpha A \beta$$

We can inductively define the set of **reachable variables**:

- **(Basis Case)** The start variable S is reachable variable.
- **(Induction Case)** If $A \in V$ is a reachable variable and $A \to \alpha \in R$, then all variables in α are reachable variables.

Eliminating Useless Variables – Example

After eliminating ϵ-productions and unit productions:

$$
\begin{aligned}
S &\to 0ABC \mid 0BC \mid 0AB \mid 0B \mid 1B \mid BB \\
A &\to ABB0 \mid BB0 \mid CC \\
B &\to 0B \mid 1 \\
C &\to CC \\
D &\to 1D \mid 1 \mid AA \mid ABB0 \mid BB0 \mid CC
\end{aligned}
$$

1. Find all **generating variables**: $\{S, A, B, D\}$ – C is non-generating.
2. Find all **reachable variables**: $\{S, A, B, C\}$ – D is unreachable.
3. Construct a new CFG by **removing** all productions that contain non-generating variables or come from unreachable variables.

$$
\begin{aligned}
S &\to 0AB \mid 0B \mid 1B \mid BB \\
A &\to ABB0 \mid BB0 \\
B &\to 0B \mid 1
\end{aligned}
$$

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Putting CFG in CNF

Our goal is to put a CFG in **Chomsky normal form (CNF)** consisting of:

$$A \to BC \qquad \text{OR} \qquad A \to a$$

where $B \neq S$ and $C \neq S$. And $S \to \epsilon$ is allowed only if $\epsilon \in L(G)$.

We can put a CFG in CNF by following the steps below:

1. If S on RHSs, add a new start variable S' and a production $S' \to S$.
2. Eliminate ϵ-productions, unit productions, and useless variables.
3. Rewrite all RHSs whose length > 1 to contain only variables: if a symbol a appears in the RHS, replace it with a new variable A and introduce a new production rule $A \to a$.
4. Replace all RHSs whose length is greater than 2 with a chain of variables. To do so, if $A \to X_1 X_2 \cdots X_n$ is a production with $n > 2$, then replace it with a sequence of productions:

$$A \to X_1 A_1 \qquad A_1 \to X_2 A_2 \qquad \cdots \qquad A_{n-2} \to X_{n-1} X_n$$

5. If ϵ is in the original language, add a production $S \to \epsilon$ (or $S' \to \epsilon$).

Putting CFG in CNF – Example 1

Let's put the following CFG in CNF:

$$
\begin{aligned}
S &\to 0ABC \mid 1B \mid BB \\
A &\to ABB0 \mid C \\
B &\to 0B \mid 1 \\
C &\to CC \mid \epsilon \\
D &\to 1D \mid AA
\end{aligned}
$$

1. If S on RHSs, add a new start variable S' and a production $S' \to S$.
2. Eliminate ϵ-productions, unit productions, and useless variables:

$$
\begin{aligned}
S &\to 0AB \mid 0B \mid 1B \mid BB \\
A &\to ABB0 \mid BB0 \\
B &\to 0B \mid 1
\end{aligned}
$$

Putting CFG in CNF – Example 1

$$
\begin{aligned}
S &\to 0AB \mid 0B \mid 1B \mid BB \\
A &\to ABB0 \mid BB0 \\
B &\to 0B \mid 1
\end{aligned}
$$

3. Rewrite all RHSs whose length > 1 to contain only variables:

$$
\begin{aligned}
S &\to XAB \mid XB \mid YB \mid BB & X &\to 0 \\
A &\to ABBX \mid BBX & Y &\to 1 \\
B &\to XB \mid 1
\end{aligned}
$$

4. Replace all RHSs whose length > 2 with a chain of variables:

$$
\begin{aligned}
S &\to XS_1 \mid XB \mid YB \mid BB & A &\to AA_1 \mid BA_2 & B &\to XB \mid 1 \\
S_1 &\to AB & A_1 &\to BA_2 & X &\to 0 \\
& & A_2 &\to BX & Y &\to 1
\end{aligned}
$$

5. If ϵ is in the original language, add a production $S \to \epsilon$: **No.**

Putting CFG in CNF – Example 2

Let's put the following CFG in CNF:

$$S \to aSb \mid \epsilon$$

1. If S on RHSs, add a new start variable S' and a production $S' \to S$.

$$S' \to S \qquad S \to aSb \mid \epsilon$$

2. Eliminate ϵ-productions, unit productions, and useless variables:

$$S' \to aSb \mid ab \qquad S \to aSb \mid ab$$

3. Rewrite all RHSs whose length > 1 to contain only variables:

$$S' \to ASB \mid AB \qquad S \to ASB \mid AB \qquad A \to a \qquad B \to b$$

4. Replace all RHSs whose length > 2 with a chain of variables:

$$S' \to AS_1 \mid AB \quad S \to AS_1 \mid AB \quad S_1 \to SB \quad A \to a \quad B \to b$$

5. If ϵ is in the original language, add a production $S' \to \epsilon$: **Yes.**

$$S' \to \epsilon \mid AS_1 \mid AB \quad S \to AS_1 \mid AB \quad S_1 \to SB \quad A \to a \quad B \to b$$

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Next Lecture

- Properties of Context-Free Languages

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